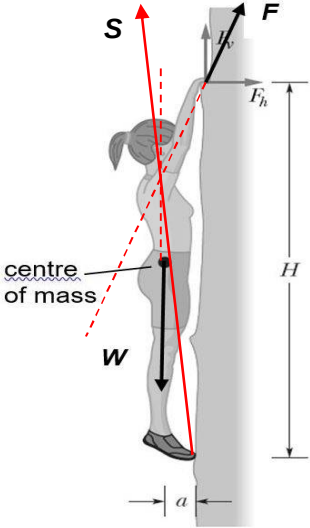


2(a)	1. The vector sum of all external forces acting on a body must be zero . 2. The vector sum of all external torques acting on a body must be zero .	[1] [1]

2(b)(i)	Taking moments about contact point between feet and wall, $\text{Weight} \times a = F_h \times H$ $(70)(9.81)(0.20) = F_h(2.0)$ $F_h = 68.67 \text{ N} = \mathbf{69 \text{ N}}$	[1] sub [1] ans
(ii)	Consider the vertical forces, $\text{Weight} = F_v + 32$ $(70)(9.81) - 32 = F_v$ $F_v = 654.7 \text{ N} = \mathbf{650 \text{ N}}$	[1] sub [1] ans
(iii)	 The diagram shows a person standing on a horizontal surface and leaning against a vertical wall. The person's feet are at a horizontal distance a from the wall. The person's center of mass is at a height H from the feet. A vertical force W acts downwards from the center of mass. A reaction force F acts upwards and to the left from the point of contact with the wall, with its horizontal component F_h and vertical component F_v shown. A dashed line S represents the line of action of the weight W, passing through the center of mass and the point of contact with the wall.	[1] correct general direction of S [1] passing through the point of inter- section between lines of action of F and W
3(a)(i)	Using Principle of Conservation of Linear Momentum	